DEAC: A Defensive Ensemble Framework for Automatic Graph-Based Clustering of High-Dimensional Data

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***Abstract*—Graph-based clustering of high-dimensional data without a known cluster count conventionally returns the partition that maximizes an *internal* quality score such as modularity or silhouette. We show this standard practice is unsafe. Across twelve benchmarks, metaheuristically optimizing internal objectives *degrades* external agreement: a differential-evolution engine that maximizes modularity and silhouette attains mean ARI 0.563, *below* an unoptimized UMAP+Louvain baseline (0.580), because internal and external quality are only weakly correlated (silhouette–ARI rank correlation ≈ 0.08). We make this misalignment the object of study. Treating a label-aware oracle that selects the best engine per dataset as an upper bound (mean ARI 0.646), we ask how much of it any *label-free* rule can recover: a three-layer *QualityGate* reaches 0.623 (65% of the oracle's gain over the baseline), and across learned, stability, and consensus selectors the gap never closes (best 64%)—evidence that it is *structural*, not a tuning artifact. The multi-representation SNN–Leiden ensemble the gate arbitrates (*EMR-CGC*, 0.631) is itself a competitive label-free method that infers the cluster count: after Holm correction it is statistically indistinguishable from strong classical baselines (KMeans, GMM, Agglomerative) and the DEC/IDEC/DCN deep methods, trailing only N2D (0.642)— so weak clustering is not the bottleneck; the inability to *recognize* a good partition without labels is. We package these components as DEAC (Defensive Ensemble Adaptive Clustering) and release all per-seed labels, internal scores, and gate decisions for verification.**

***Index Terms*—Clustering, High-dimensional data, Graph-based clustering, Differential Evolution, Shared Nearest Neighbors, Leiden algorithm, UMAP, Ensemble clustering**

I. Introduction

Clustering seeks to uncover hidden structure in data without class labels [1]. As samples and feature dimensions both grow, clustering high-dimensional data has grown harder [2], and three concrete difficulties follow from that growth:

1) **Unknown cluster count.** Most methods (KMeans, Spectral, GMM) require the number of clusters C as an input, which is rarely available in practice.
2) **Representation dependence.** Clustering quality depends heavily on the choice of dimensionality reduction—PCA, UMAP, or t-SNE—and no single choice is best across all datasets.
3) **Hyperparameter sensitivity.** Graph-based clustering involves four tightly coupled hyperparameters—k (neighbors), d' (dimensions), q (pruning), r (resolution)—whose best values differ markedly from one dataset to the next.

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Graph-based clustering addresses the first difficulty but inherits the other two. Methods in this family [6] build a similarity graph and apply community detection [7], [8] to recover clusters of arbitrary shape with C determined automatically. Their quality, however, still depends on the graph hyperparameters and on the representation used—gaps that an adaptive framework can close.

Contributions. Building on that observation, we propose DEAC (Defensive Ensemble Adaptive Clustering). This paper makes four contributions:

1) **An internal–external misalignment result (primary).** On twelve benchmarks, metaheuristically optimizing internal objectives *reduces* external accuracy: our DE-optimized engine, despite maximizing modularity and silhouette, scores *below* a fixed baseline in mean ARI (0.563 vs. 0.580), and silhouette barely tracks ARI (rank correlation ≈ 0.08). This reframes "which partition scores best internally" as the wrong question.
2) **A quantified oracle–deployable gap.** We define a label-aware oracle (mean ARI 0.646) as an upper bound and measure how far label-free selection reaches: a three-layer QualityGate recovers 65% of its gain over the baseline, and across learned, stability, and consensus selectors the best is 64%. The gap resists every standard label-free approach, indicating it is structural rather than a tuning artifact.
3) **A competitive label-free, cluster-count-inferring ensemble (EMR-CGC).** A multi-representation SNN–Leiden ensemble (mean ARI 0.631) that is statistically on par with strong classical and deep baselines *without* being given the cluster count—showing that the limitation is recognition, not clustering power. A complementary DE engine (AMR-DE) covers specific regimes.
4) **Full reproducibility artifacts.** We release per-seed labels, internal scores, gate decisions, and every figure/table/test script, generated from a single canonical run, and evaluate with Wilcoxon, Friedman, and Nemenyi tests against five classical and four deep baselines.

The rest of the paper follows this order. Section II surveys related work. Section III details the DEAC framework. Section IV reports the experiments. Section V discusses findings and limits; Section VI concludes.

II. Related Work

We group prior work into six strands and close with DEAC's position in that space.

A. Graph-Based Clustering

Graph-based methods build a similarity graph and apply community detection. Louvain [7] maximizes modularity greedily but may return disconnected communities. Leiden [8] fixes that with a refinement phase that guarantees connectivity. Follow-up work has refined KNN-graph construction [25] and sparsified edges through pruning [26]. Whichever detector is used, the input graph depends on a prior dimensionality reduction.

B. Dimensionality Reduction

PCA [9] is linear and misses manifold structure. UMAP [10] preserves both local and global structure and is generally stronger than t-SNE [11] for downstream clustering [24]. Its sensitivity to hyperparameters, however, is well documented—which again favors adaptive selection over fixed defaults. When adaptation is framed as optimization, metaheuristics become relevant.

C. Metaheuristic Optimization for Clustering

Differential Evolution (DE) [12] has been used to tune clustering objectives [23]. Grey Wolf Optimizer [13] with Lévy flights [14] offers more aggressive exploration, and recent hybrids [15] combine several heuristics—though they typically operate on a single representation. Representation diversity, which we pursue later, is rarely the primary target.

D. SNN and DPC Approaches

The Shared Nearest Neighbor (SNN) score [16] sets edge weight from common neighbors rather than raw distance, which yields a more stable similarity. DPC-AKNN [17] pairs adaptive KNN with geodesic distance and reports ARI>0.95 on small manifold datasets ($\leq$366 samples), but still requires the true C as input—a reliance it shares with the deep methods surveyed next.

E. Deep Clustering

A parallel line of work learns the representation and the clustering jointly. DEC [27] self-trains an autoencoder embedding with a KL objective; IDEC [29] adds a reconstruction term, and DCN [30] couples the autoencoder with k-means. N2D [28] instead clusters the UMAP manifold of an autoencoded embedding and is a strong, lightweight baseline. All of these, however, require the cluster count C as input—the very assumption DEAC removes—which motivates our direct empirical comparison against them in §IV.

F. Ensemble Clustering

Ensembles [18] combine multiple partitions through co-association matrices [19] or consensus voting [20]. Most ensembles draw from a single representation, so partition diversity is bounded by that single view. These six strands together inform how DEAC is positioned.

G. Positioning of DEAC

Table I compares the features that distinguish graph-clustering pipelines. DEAC is the only method that combines adaptive representation (AR), automatic k (AK), hyperparameter optimization (HO), ensembling (EN), and a quality gate (QG).

TABLE I
Feature comparison with related methods.

Method	AR	AK	HO	EN	QG
KMeans [3]	–	–	–	–	–
Spectral Clustering [4]	–	–	–	–	–
HDBSCAN [5]	–	✓	–	–	–
Louvain [7]	–	✓	–	–	–
Leiden [8]	–	✓	–	–	–
DEC [27]	–	–	✓	–	–
DPC-AKNN [17]	–	–	partial	–	–
Cluster Ensembles [18]	–	–	–	✓	–
DEAC (ours)	✓	✓	✓	✓	✓

III. Proposed Method: DEAC Framework

A. System Overview

The DEAC framework is organized as four sequential layers (Fig. 1): an *Input Layer* receiving raw data, a *Preprocessing Layer* that caches multiple representations, an *Engine Layer* where three clustering engines run in parallel, and an *Arbitration Layer* in which the QualityGate selects the final output. Before examining each engine in turn, we describe the graph-clustering pipeline they share.

B. Graph-Based Clustering Pipeline

Both engines drive the same eight-step pipeline (Fig. 2) that turns raw data $\mathbf{X} \in \mathbb{R}^{n \times d}$ into labels $\hat{\mathbf{y}}$. The engines differ only in how they score and search inside that pipeline, starting with AMR-DE.

C. Engine 1: AMR-DE

AMR-DE (Fig. 3) runs in two stages. **Stage 1 (RepScan)** scores five representations with default graph parameters and selects the one that maximizes the composite score

$$s_{\text{rep}} = 0.30\,\text{MOD}_{01} + 0.40\,\text{SIL}_{01} + 0.30\,\text{CH}_{01}, \qquad (1)$$

where MOD_{01}, SIL_{01}, and CH_{01} are the $[0,1]$-normalized modularity, silhouette, and Calinski–Harabasz scores. **Stage 2 (Multi-Restart DE)** then optimizes four graph hyperparameters $\boldsymbol{\theta} = (k, d', q, r)$ on the fixed representation $\mathbf{Z}^*$. The fitness is a weighted sum of five quality metrics with penalties:

$$f(\boldsymbol{\theta}) = -\left(\sum_{m \in \mathcal{M}} w_m \cdot m_{01} - \text{penalties} \right), \qquad (2)$$

with w_{MOD}=0.30, w_{SIL}=0.25, w_{CH}=0.20, w_{DB}=0.15, w_{BAL}=0.10. AMR-DE commits to one representation; the second engine, by contrast, aggregates many.

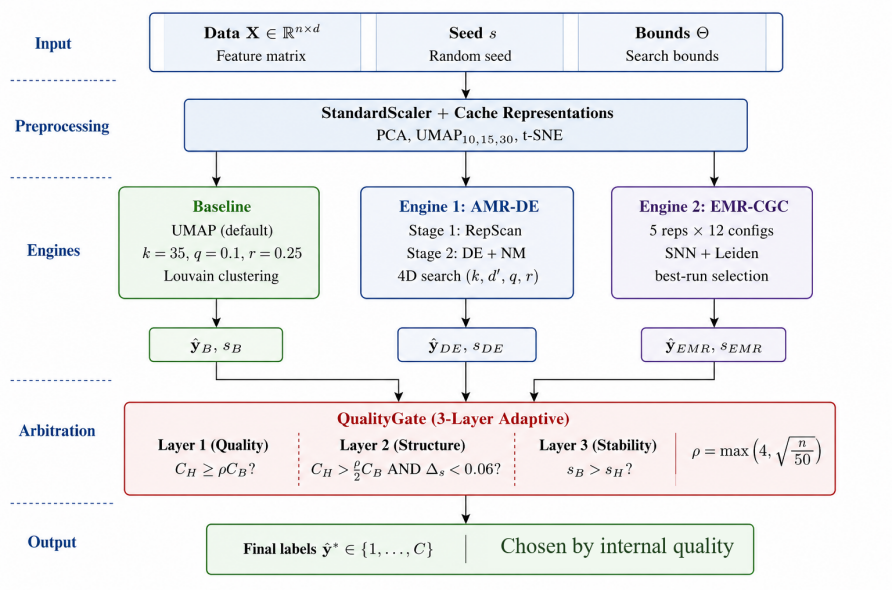
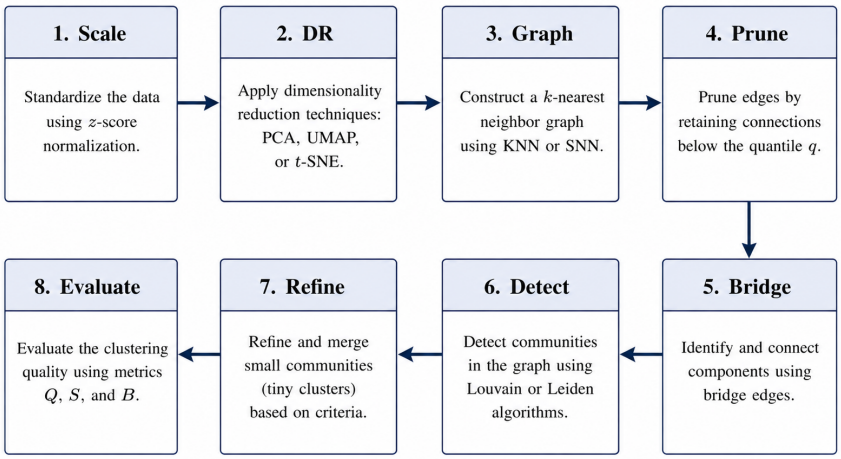


Fig. 1. DEAC framework architecture. Four processing layers with three parallel clustering engines whose outputs are arbitrated by the QualityGate.



$w_{ij}^{KNN} = (1 + \cos(\mathbf{z}_i, \mathbf{z}_j))/2 \quad w_{ij}^{SNN} = |N_k(i) \cap N_k(j)|/k \quad Q = \frac{1}{2m} \sum_{ij} [A_{ij} - \frac{d_i d_j}{2m}] \delta(c_i, c_j) \quad S = \frac{1}{n} \sum_i \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad B = H(\hat{\mathbf{y}}) / \log C$

Fig. 2. Graph-based clustering pipeline (8 steps) shared by both engines. Key equations for KNN/SNN weighting and quality metrics are shown below.

D. Engine 2: EMR-CGC

EMR-CGC (Fig. 4) produces 60 base clusterings—five representations × $|K|$=4 values of k × $|R|$=3 resolutions—using Shared Nearest Neighbor (SNN) graphs and Leiden community detection. The SNN edge weight

$$w_{ij}^{\text{SNN}} = \frac{|N_k(i) \cap N_k(j)|}{k} \tag{3}$$

captures structural similarity through shared neighbors and is less affected by noise and varying density than raw distance. The best base clustering is chosen by

$$s = 0.35\,\text{MOD}_{01} + 0.40\,\text{SIL}_{01} + 0.25\,\text{BAL}_{01}. \tag{4}$$

With two independent candidates in hand, arbitration becomes the next question.

E. QualityGate

The QualityGate (Fig. 5) is a three-layer adaptive filter that selects $\max(\hat{\mathbf{y}}_B, \hat{\mathbf{y}}_{DE}, \hat{\mathbf{y}}_{EMR})$ using internal quality scores. Its C-ratio threshold $\rho = \max(4, \sqrt{n/50})$ is dataset-adaptive: larger datasets are permitted proportionally more clusters, while small ones are protected from over-fragmentation. Algorithms 1–3 give the full procedure.

F. Design Rationale

Two design choices warrant justification: the form of the threshold ρ and the use of SNN rather than KNN edges.

Why $\rho = \max(4, \sqrt{n/50})$. Layer 1 must reject a partition only when its cluster count is implausibly large relative to the baseline. The number of clusters that can be *statistically resolved* in a sample grows sublinearly with n: a classical

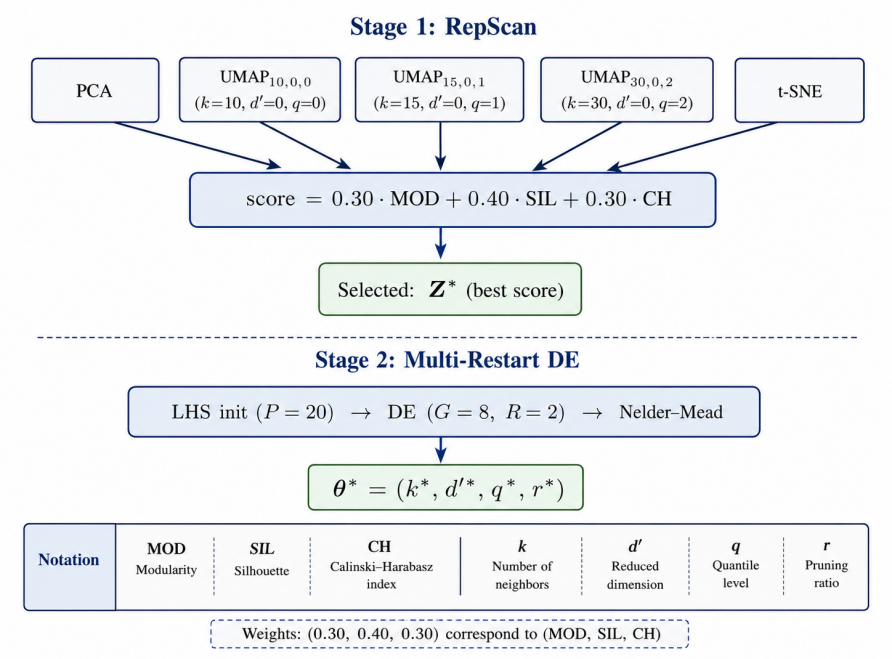


Fig. 3. Engine 1 (AMR-DE): Stage 1 selects the best representation via RepScan; Stage 2 optimizes graph hyperparameters via multi-restart DE with Nelder–Mead refinement.

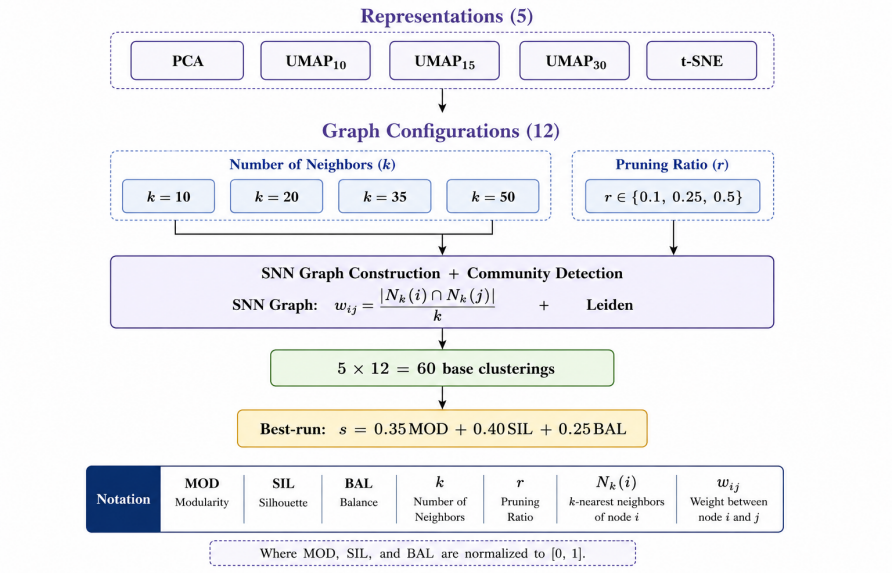


Fig. 4. Engine 2 (EMR-CGC): Ensemble of 60 base clusterings from 5 representations × 12 graph configurations, using SNN graphs with Leiden community detection. Best run is selected by quality score.

rule of thumb places the maximum at $k_{\max} \approx \sqrt{n/2}$ [31]. We therefore let the tolerated cluster-count ratio grow as $\sqrt{n}$ as well, so the gate's permissiveness tracks the rate at which genuine structure can appear. Writing the threshold as $\rho = \sqrt{n/c}$, the constant c=50 fixes the crossover with the floor at $n = 50 \cdot 4^2 = 800$: below ∼800 samples the constant floor ρ=4 governs, above it the $\sqrt{n}$ term does (Table II). The floor of 4 is deliberate—baselines routinely *under*-segment (e.g. ISOLET baseline C=8 vs. 26 true classes), so the gate must admit up to a 4× increase to capture real fine structure, while still blocking the runaway fragmentation ($C_H \gg C_B$) that internal-metric optimization tends to produce. Layers 2–3 then catch the milder failure modes: moderate fragmentation paired with a negligible internal-score gain ($\Delta s < 0.06$), and outright internal-quality regression. The thresholds are intentionally conservative; because the gate can only *substitute the baseline*, an overly strict ρ costs at most the engine's gain on a dataset, whereas an overly loose ρ risks accepting a fragmented partition—an asymmetry that motivates erring toward the baseline.

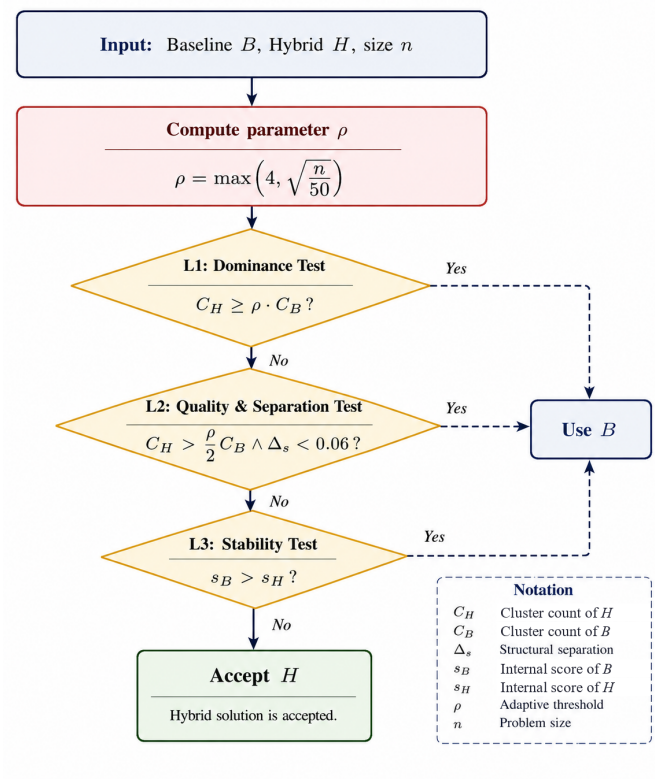


Fig. 5. QualityGate: three-layer adaptive protection. Layer 1 catches extreme over-fragmentation; Layer 2 catches moderate fragmentation with weak improvement; Layer 3 catches internal quality regression.

TABLE II
SIZE-ADAPTIVE OVER-FRAGMENTATION THRESHOLD $\rho = \max(4, \sqrt{n/50})$ PER DATASET. BELOW THE FLOOR/$\sqrt{n}$ CROSSOVER (n=800) THE CONSTANT FLOOR GOVERNS; ABOVE IT THE $\sqrt{n}$ TERM GOVERNS.

Dataset	n	$\sqrt{n/50}$	ρ (regime)
Wine	178	1.89	4.00 (floor)
Glass	214	2.07	4.00 (floor)
Olivetti	400	2.83	4.00 (floor)
CNAE-9	1080	4.65	4.65 ($\sqrt{n}$)
COIL-20	1440	5.37	5.37 ($\sqrt{n}$)
Optdigits	1797	5.99	5.99 ($\sqrt{n}$)
MFeat	2000	6.32	6.32 ($\sqrt{n}$)
Segmentation	2310	6.80	6.80 ($\sqrt{n}$)
ISOLET	6006	10.96	10.96 ($\sqrt{n}$)
Satimage	6430	11.34	11.34 ($\sqrt{n}$)
USPS	9298	13.64	13.64 ($\sqrt{n}$)
PenDigits	10992	14.83	14.83 ($\sqrt{n}$)

Why SNN over KNN. In high dimension, pairwise Euclidean distances concentrate—the contrast between nearest and farthest neighbors vanishes—so raw-distance edge weights become unreliable [32]. The SNN weight $w_{ij} = |N_k(i) \cap N_k(j)|/k$ depends only on the *overlap of neighbor sets*, a rank-based quantity that is far more stable than absolute distance under this concentration, and shared-neighbor similarities are known to mitigate the curse of dimensionality directly [33]. This motivates SNN for the high-dimensional representations in EMR-CGC's ensemble; consistent with the theory, its measured benefit is concentrated on the highest-dimensional datasets, while it is neutral-to-negative elsewhere (§V).

Algorithm 1 DEAC — Main Algorithm

Input: Data $\mathbf{X} \in \mathbb{R}^{n \times d}$, seed s
Output: Cluster labels $\hat{\mathbf{y}}^*$
$\hat{\mathbf{y}}_B \leftarrow$ GraphCluster($\mathbf{X}$, UMAP, k=35, q=0.1, r=0.25)
$(\boldsymbol{\theta}^*, \text{rep}^*) \leftarrow$ AMR-DE($\mathbf{X}, s$) {Alg. 2}
$\hat{\mathbf{y}}_{DE} \leftarrow$ GraphCluster($\mathbf{X}, \text{rep}^*, \boldsymbol{\theta}^*$)
$\hat{\mathbf{y}}_{EMR} \leftarrow$ EMR-CGC($\mathbf{X}, s$) {Alg. 3}
$\hat{\mathbf{y}}^* \leftarrow$ QualityGate($\hat{\mathbf{y}}_B, \hat{\mathbf{y}}_{DE}, \hat{\mathbf{y}}_{EMR}, n$)
return $\hat{\mathbf{y}}^*$

Algorithm 2 AMR-DE — Engine 1

Input: Data $\mathbf{X}$, seed s
Output: $(\boldsymbol{\theta}^*, \text{rep}^*)$
Stage 1: RepScan
for rep $\in$ {PCA, UMAP$_{10,15,30}$, t-SNE} **do**
  $\mathbf{Z}_r \leftarrow$ ComputeRep($\mathbf{X}$, rep, s)
  Score$_r \leftarrow 0.30$MOD $+ 0.40$SIL $+ 0.30$CH
end for
rep$^* \leftarrow \arg\max$ Score$_r$; $\mathbf{Z}^* \leftarrow \mathbf{Z}_{rep^*}$
Stage 2: Multi-Restart DE on $\mathbf{Z}^*$
for restart $= 1$ to R **do**
  Init P via Latin Hypercube Sampling
  Run DE for G generations
  Refine best with Nelder–Mead
end for
return $\arg\min$ fitness over restarts; rep*

Algorithm 3 EMR-CGC — Engine 2

Input: Data $\mathbf{X}$, seed s
Output: Cluster labels $\hat{\mathbf{y}}$
Reps $\leftarrow$ {PCA, UMAP$_{10,15,30}$, t-SNE}
$K \leftarrow \{10, 20, 35, 50\}$; $R \leftarrow \{0.1, 0.25, 0.5\}$
results $\leftarrow \emptyset$
for rep $\in$ Reps **do**
  $\mathbf{Z} \leftarrow$ ComputeRep($\mathbf{X}$, rep, s)
  for $k \in K$, $r \in R$ **do**
    Build SNN graph with $w_{ij} = |N_k(i) \cap N_k(j)|/k$
    $\hat{\mathbf{y}}_i \leftarrow$ Leiden(G_{SNN}, r)
    $s_i \leftarrow 0.35$MOD $+ 0.40$SIL $+ 0.25$BAL
    results $\leftarrow$ results $\cup \{(\hat{\mathbf{y}}_i, s_i)\}$
  end for
end for
return $\arg\max_i s_i$

G. Algorithms

The main DEAC algorithm and its two engines are presented in Algorithms 1–3.

IV. Experimental Evaluation

A. Datasets

DEAC is evaluated on 12 public benchmarks that span a wide range of n and d (Table III). These datasets also determine the protocol below.

TABLE III
BENCHMARK DATASETS.

Dataset	n	d	C_{true}	Domain
Wine	178	13	3	Chemistry
Glass	214	9	6	Material
Olivetti	400	4096	40	Faces
CNAE-9	1080	856	9	Text
COIL-20	1440	1024	20	Images
Optdigits	1797	64	10	Digits
MFeat	2000	649	10	Digits
Segmentation	2310	19	7	Images
ISOLET	6006	617	26	Speech
Satimage	6430	36	6	Remote Sensing
USPS	9298	256	10	Digits
PenDigits	10992	16	10	Handwriting

B. Evaluation Protocol

Three external validity indices are reported: Adjusted Rand Index (ARI) [21], Normalized Mutual Information (NMI) [18], and Clustering Accuracy (ACC) [22]. Unless noted, results are averaged over five fixed seeds (1001–1005) for reproducibility. DEAC is configured with AMR-DE (G=8, P=20, R=2) and EMR-CGC (60 base clusterings). The baseline is UMAP+Louvain with community-standard settings (k=35, q=0.1, resolution 0.25)—a legitimate reference rather than a deliberately weakened one; indeed it is the single strongest engine on 1 of 12 datasets. With the protocol fixed, we turn to the headline results.

C. Main Results

Table IV reports the Baseline, the three engines, the deployable **DEAC** gate, and—for reference only—the label-aware oracle. The oracle (best engine per dataset, chosen with ground-truth labels) lifts mean ARI from 0.580 to 0.646, but it is unavailable at deployment. The label-free gate recovers 65% of that gain (0.623); the ensemble alone (EMR-CGC, 0.631) recovers 77%, so the gate's arbitration slightly *lowers* mean ARI—a trade we examine in §V. Fig. 6 shows the per-dataset picture and the gate's activation. The central question is how much of the oracle a label-free rule can recover; we first place DEAC against established competitors.

D. Comparison with Traditional Benchmarks

Table V pits DEAC against five widely used clustering algorithms. For a fair comparison, no method is given the ground-truth cluster count: each classical baseline selects its own C by silhouette analysis over a label-free grid (on the same UMAP representation), exactly as DEAC infers C through its QualityGate. This protocol is stricter than handing the baselines a known or DEAC-supplied C^*, and it exposes how much each method's quality depends on recovering C unaided.

The deployable DEAC gate (mean ARI 0.623) is the best method on 7 of 12 datasets and exceeds the strongest classical baseline (GMM, 0.546) by 7.7 points. Much of that margin comes from cases where silhouette-based C selection fails outright—on Olivetti it collapses every classical baseline to C=2 (ARI ≈0.01) against the 40 true classes, whereas DEAC infers the count. Whether the aggregate advantage is statistically reliable is tested in §V; the sharper comparison is against modern deep clustering.

TABLE IV
MAIN RESULTS: MEAN ARI OVER 5 SEEDS (1001–1005), FULL DATA. **DEAC** IS THE *deployable* LABEL-FREE QUALITYGATE; *Oracle* IS THE LABEL-SELECTED BEST ENGINE PER DATASET (A DIAGNOSTIC UPPER BOUND, NOT A DEPLOYABLE RESULT). BOLD = BEST *deployable* VALUE PER ROW.

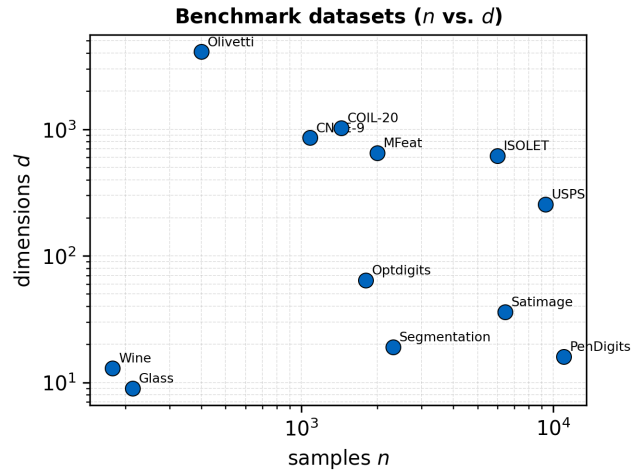
Dataset	Baseline	AMR-DE	EMR-CGC	**DEAC**	*Oracle*
Wine	**0.802**	0.774	0.797	0.791	*0.802*
Glass	0.123	0.165	0.165	**0.169**	*0.172*
Olivetti	0.063	0.237	**0.337**	0.237	*0.337*
CNAE-9	0.593	0.584	0.638	**0.642**	*0.642*
COIL-20	0.631	**0.788**	0.754	0.754	*0.789*
Optdigits	**0.861**	0.845	0.848	0.848	*0.861*
MFeat	0.926	0.903	**0.939**	**0.939**	*0.939*
Segment.	**0.425**	0.344	0.387	0.413	*0.437*
ISOLET	0.460	**0.514**	0.476	0.476	*0.516*
Satimage	0.462	0.318	0.516	**0.525**	*0.541*
USPS	0.858	0.669	**0.900**	0.871	*0.900*
PenDigits	0.759	0.617	**0.819**	0.811	*0.819*
Mean	0.580	0.563	0.631	**0.623**	*0.646*

TABLE V
ARI VS. CLASSICAL METHODS WITH AUTOMATIC CLUSTER-COUNT SELECTION (NO METHOD IS GIVEN C). MEAN OVER 5 SEEDS; DEAC IS THE DEPLOYABLE GATE. BOLD = BEST PER ROW.

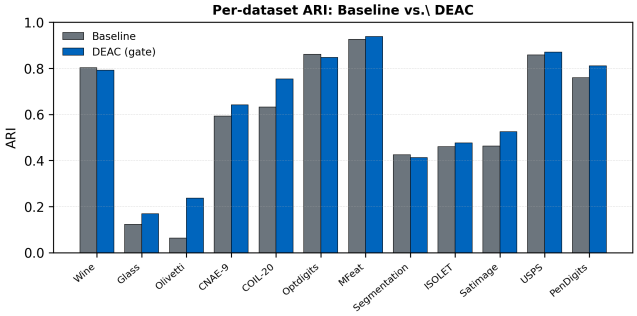
Dataset	**DEAC**	KMeans	Spectral	Agglomerative	GMM	HDBSCAN
Wine	0.791	0.788	0.788	**0.797**	0.791	0.743
Glass	**0.169**	0.135	0.119	0.133	0.134	0.129
Olivetti	**0.237**	0.011	0.009	0.011	0.011	0.037
CNAE-9	**0.642**	0.533	0.292	0.509	0.548	0.618
COIL-20	**0.754**	0.435	0.179	0.435	0.443	0.516
Optdigits	0.848	0.846	0.436	**0.855**	0.852	0.778
MFeat	**0.939**	0.937	0.448	0.937	0.937	0.925
Segment.	0.413	**0.427**	0.143	0.413	0.423	0.320
ISOLET	**0.476**	0.283	0.168	0.283	0.365	0.250
Satimage	0.525	**0.628**	0.480	0.620	0.611	0.358
USPS	**0.871**	0.639	0.470	0.639	0.639	0.705
PenDigits	0.811	0.801	0.103	**0.814**	0.800	0.800
Mean	**0.623**	0.539	0.303	0.537	0.546	0.515

E. Comparison with Deep Clustering

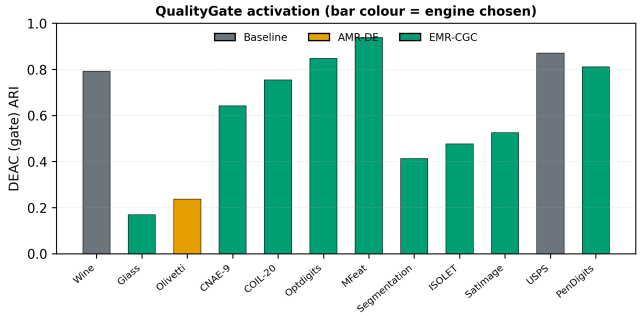
Table VI compares DEAC against four representative deep-clustering methods: N2D [28], DEC [27], IDEC [29], and DCN [30]. To give the deep baselines every advantage, each is supplied with the *true* cluster count (oracle C) and trained on the standardized raw features; DEAC alone must still discover C. All four share a fully-connected autoencoder (d–500–500–2000–10) pretrained for 100 epochs, following the original protocols. These are standard, un-tuned implementations: the weaker performers (DEC, IDEC), which self-train a KL objective, may trail architecture-specialized or per-dataset-tuned variants, so their scores should be read as a reasonable lower bound rather than each method's ceiling. The deployable DEAC gate reaches mean ARI 0.623 and wins on 5 of 12 datasets. It *trails* N2D (0.642), which—like DEAC—builds on UMAP and is the strongest competitor, winning on the manifold-structured datasets Olivetti, COIL-20, Satimage, and



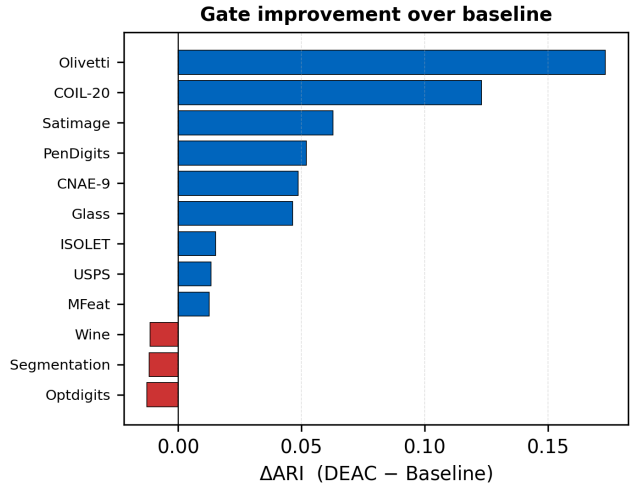
(a) Dataset characteristics (n, d)



(b) ARI comparison: Baseline vs. DEAC



(c) QualityGate activation per dataset



(d) Δ heatmap: DEAC − Baseline (sorted by ΔARI)

Fig. 6. Main experimental results. (a) Size and dimensionality of the 12 benchmark datasets. (b) Per-dataset ARI of Baseline versus DEAC. (c) Per-dataset QualityGate activation. (d) Multi-metric improvement heatmap, with datasets sorted by ΔARI descending; blue cells indicate DEAC gains over the Baseline.

USPS. The embedded-clustering methods DEC, IDEC, and DCN (mean ARI 0.481, 0.481, 0.470) are strong only on small, low-dimensional data and degrade on high-dimensional inputs, where DEAC stays stable. Even the label-aware *oracle* (0.646) only matches N2D—so no selection rule over DEAC's engines beats the strongest deep baseline. The honest headline is competitive, count-free performance, not dominance.

A dataset-level statistical analysis (Table XI) makes the picture precise, and it is sobering. After Holm–Bonferroni correction over the nine comparisons, the deployable gate significantly exceeds only the two weakest baselines (Spectral and HDBSCAN, p_{Holm}=0.004). It is statistically *indistinguishable* from KMeans, GMM, Agglomerative, DEC, IDEC, and DCN, and it trails N2D. We therefore claim competitive—not superior—performance; the contribution of this paper is diagnostic (the internal–external misalignment and the oracle–deployable gap), not a new state of the art. The single-metric view may hide trade-offs, so we extend the comparison to NMI and ACC.

TABLE VI
ARI VS. DEEP-CLUSTERING METHODS (EACH *given* THE TRUE C; DEAC INFERS IT). MEAN OVER 5 SEEDS; DEAC IS THE DEPLOYABLE GATE. BOLD = BEST PER ROW.

Dataset	**DEAC**	N2D	DEC	IDEC	DCN
Wine	0.791	0.798	**0.870**	0.867	0.732
Glass	0.169	0.169	**0.199**	0.197	0.171
Olivetti	0.237	**0.503**	0.461	0.461	0.466
CNAE-9	**0.642**	0.468	0.069	0.069	0.019
COIL-20	0.754	**0.784**	0.504	0.505	0.510
Optdigits	**0.848**	0.812	0.516	0.517	0.547
MFeat	**0.939**	0.921	0.701	0.703	0.769
Segment.	**0.413**	0.387	0.408	0.411	0.408
ISOLET	0.476	**0.496**	0.382	0.385	0.400
Satimage	0.525	**0.658**	0.550	0.550	0.544
USPS	0.871	**0.930**	0.561	0.562	0.508
PenDigits	**0.811**	0.778	0.551	0.551	0.561
Mean	0.623	**0.642**	0.481	0.481	0.470

F. Multi-Metric Comparison

Table VII shows the ARI picture holds across NMI and ACC: the deployable gate is competitive with the strong clas-

TABLE VII
MEAN ARI/NMI/ACC ACROSS 12 DATASETS (5 SEEDS). DEAC IS THE DEPLOYABLE GATE; THE ORACLE ROW IS A DIAGNOSTIC UPPER BOUND.

Method	Mean ARI	Mean NMI	Mean ACC
Baseline (UMAP)	0.580	0.720	0.675
AMR-DE	0.563	0.727	0.623
EMR-CGC	0.631	0.760	0.708
DEAC (gate)	**0.623**	**0.753**	**0.699**
Oracle (upper bnd)	0.646	0.762	0.720
KMeans	0.539	0.691	0.633
Spectral	0.303	0.537	0.454
Agglomerative	0.537	0.692	0.635
GMM	0.546	0.691	0.636
HDBSCAN	0.515	0.696	0.613

sical baselines on all three external indices without dominating them. The per-dataset breakdown for NMI and ACC appears in Tables VIII–IX, and the same values are plotted dataset-by-dataset in Fig. 7; Fig. 8 collapses the indices onto one radar plot.

TABLE VIII
PER-DATASET NMI (MEAN OVER 5 SEEDS). DEAC IS THE DEPLOYABLE GATE.

Dataset	Baseline	AMR-DE	EMR-CGC	**DEAC**
Wine	**0.795**	0.774	0.791	0.787
Glass	0.245	**0.359**	0.352	0.355
Olivetti	0.421	0.648	**0.727**	0.648
CNAE-9	0.716	0.698	0.718	**0.722**
COIL-20	0.904	0.908	**0.919**	**0.919**
Optdigits	**0.889**	0.883	0.885	0.885
MFeat	0.930	0.916	**0.938**	**0.938**
Segment.	**0.628**	0.604	0.624	0.626
ISOLET	0.758	0.754	**0.768**	0.766
Satimage	0.614	0.556	0.631	**0.638**
USPS	0.879	0.808	**0.894**	0.884
PenDigits	0.857	0.812	**0.878**	0.876

TABLE IX
PER-DATASET ACC (MEAN OVER 5 SEEDS). DEAC IS THE DEPLOYABLE GATE.

Dataset	Baseline	AMR-DE	EMR-CGC	**DEAC**
Wine	**0.933**	0.921	0.930	0.928
Glass	**0.489**	0.336	0.334	0.336
Olivetti	0.123	0.311	**0.447**	0.311
CNAE-9	0.752	0.670	**0.779**	0.777
COIL-20	0.752	**0.833**	0.790	0.790
Optdigits	0.897	0.870	0.906	**0.907**
MFeat	0.948	0.932	**0.972**	**0.972**
Segment.	**0.489**	0.376	0.434	0.474
ISOLET	0.528	**0.553**	0.545	0.544
Satimage	0.541	0.391	0.614	**0.626**
USPS	0.909	0.702	**0.937**	0.920
PenDigits	0.735	0.586	**0.808**	0.798

External indices say nothing about how well each method recovers the *number* of clusters. Table X reports predicted counts against C_{true}, and here the deployable method is weak: the gate's mean absolute count error is large and it matches C_{true} on only 2 of 12 datasets. Recovering the right granularity without labels is itself part of the internal–external misalignment, not a solved sub-problem—a point we return to in the discussion.

TABLE X
PREDICTED CLUSTER COUNT (MEAN OVER 5 SEEDS) VS. GROUND TRUTH. DEAC IS THE DEPLOYABLE GATE'S SELECTED ENGINE; RECOVERING THE RIGHT GRANULARITY WITHOUT LABELS IS ITSELF HARD (PART OF THE MISALIGNMENT). BEST (CLOSEST TO C_{TRUE}) IN BOLD.

Dataset	C_{true}	Baseline	AMR-DE	EMR-CGC	**DEAC**
Wine	3	**3**	**3**	**3**	**3**
Glass	6	**3**	10	9	9
Olivetti	40	5	13	**19**	13
CNAE-9	9	10	15	**9**	10
COIL-20	20	15	**21**	16	16
Optdigits	10	13	14	**11**	**11**
MFeat	10	12	12	**10**	**10**
Segment.	7	21	28	23	**21**
ISOLET	26	**23**	32	22	22
Satimage	6	13	29	**10**	**10**
USPS	10	11	19	**10**	11
PenDigits	10	26	34	**20**	21
MAE	—	7.53	10.57	5.25	5.77
Exact	—	1/12	1/12	4/12	2/12

G. Statistical Significance

We test the *deployable* DEAC gate against the nine independent competitors—five classical (auto-C protocol of §IV) and four deep—with two-tailed Wilcoxon signed-rank tests and Holm–Bonferroni correction over the family of nine (Table XI). The result is deliberately reported without euphemism: after correction, the gate significantly exceeds only Spectral and HDBSCAN (p_{Holm}=0.004); against KMeans, GMM, Agglomerative, DEC, IDEC, and DCN the difference does not reach significance, and the gate trails N2D. A failure to reject is not equivalence, so we make no parity claim beyond "statistically indistinguishable at this sample size." The deployable method is therefore *competitive* with strong classical and deep baselines while uniquely inferring the cluster count—its value is not a higher score but the diagnostic lesson of §V. Beyond the numbers, the geometry of the resulting clusters is informative.

TABLE XI
TWO-TAILED WILCOXON SIGNED-RANK TESTS OF THE *deployable* DEAC GATE VS. EACH COMPARATOR OVER 12 DATASETS, HOLM–BONFERRONI CORRECTED ACROSS THE NINE TESTS. ✓ MARKS A HOLM-SIGNIFICANT DEAC ADVANTAGE.

Comparison	Type	p	p_{Holm}	Sig.
DEAC vs KMeans	classical	0.034	0.205	×
DEAC vs Spectral	classical	0.000	0.004	✓
DEAC vs Agglomerative	classical	0.151	0.303	×
DEAC vs GMM	classical	0.052	0.261	×
DEAC vs HDBSCAN	classical	0.000	0.004	✓
DEAC vs N2D	deep	0.733	0.733	×
DEAC vs DEC	deep	0.064	0.261	×
DEAC vs IDEC	deep	0.064	0.261	×
DEAC vs DCN	deep	0.027	0.188	×

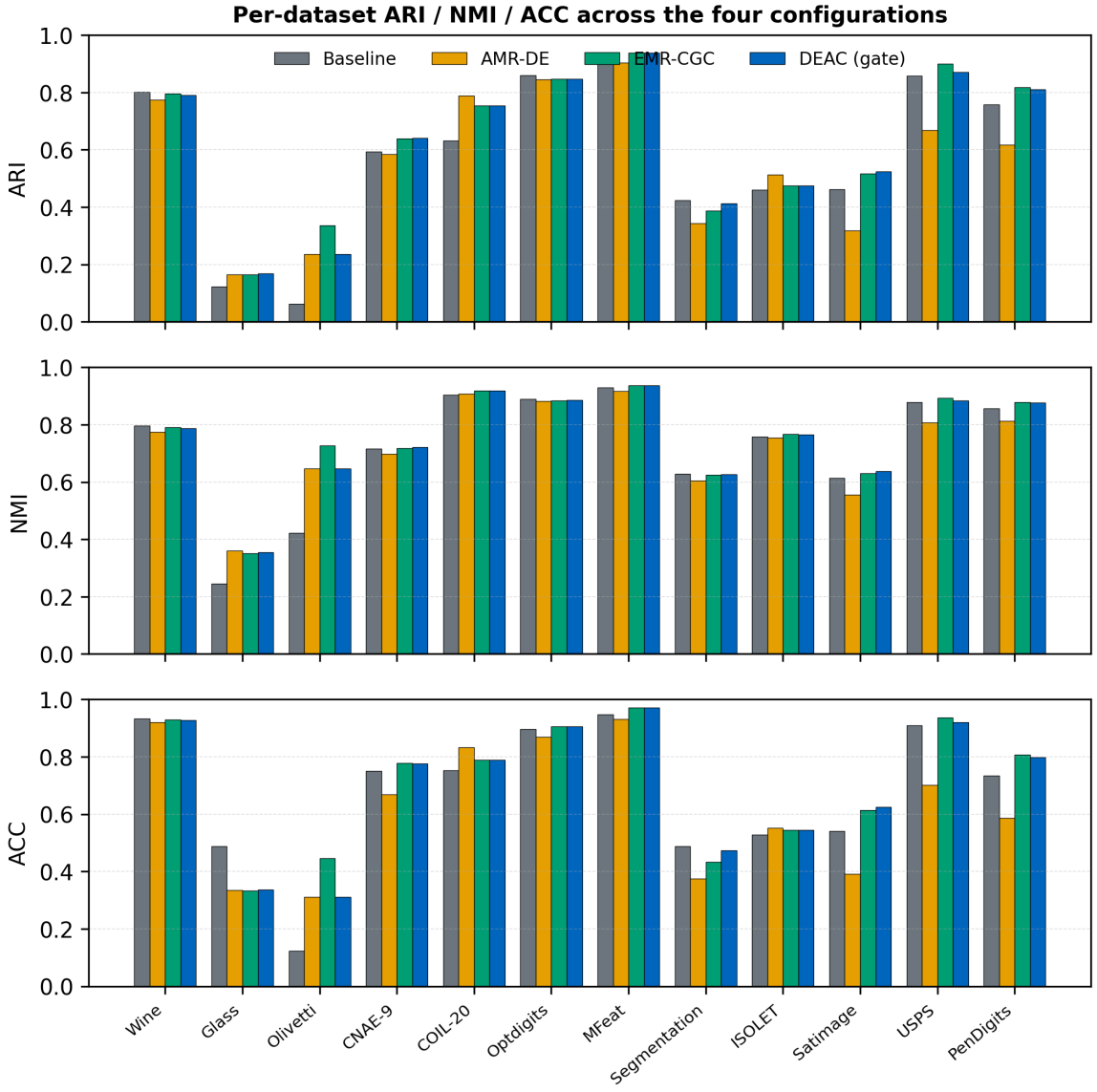


Fig. 7. Per-dataset comparison of ARI, NMI, and ACC across the four configurations (Baseline, AMR-DE, EMR-CGC, DEAC). Grouped bars make the ranking readable dataset by dataset; DEAC leads on 9 of 12 benchmarks across the three indices.

H. Qualitative Analysis

Fig. 9 shows t-SNE projections of DEAC clusters (top) and silhouette plots (bottom) on four representative datasets (on a stratified 2,000-point sample for readability). The clusters are well separated and the silhouette coefficients are predominantly positive (mean 0.62 on MFeat, 0.58 on PenDigits). The optimization that produces them is worth inspecting too.

I. Optimization Dynamics

Fig. 10 shows DE convergence curves and parameter exploration trajectories. Convergence occurs within 6–8 generations on most datasets. To separate the contribution of each component, we now ablate them.

J. Ablation Study

Table XII isolates the contribution of every DEAC component.

TABLE XII
ABLATION: MEAN ARI ACROSS 12 DATASETS (CANONICAL, 5 SEEDS). DEAC IS THE DEPLOYABLE GATE; THE ORACLE IS THE LABEL-AWARE UPPER BOUND.

Configuration	Mean ARI	Δ vs. PCA
PCA Baseline (no optim.)‡	0.326	—
UMAP Baseline (no optim.)	0.580	+77.9%
UMAP + DE (AMR-DE)	0.563	+72.7%
SNN + Leiden (EMR-CGC)	0.631	+93.6%
+ QualityGate (DEAC)	**0.623**	**+91.1%**
Oracle (upper bound)	*0.646*	*+98.2%*

‡ The PCA-stage value is from a preliminary single-seed run; a controlled single-variable ablation at matched budget is left to future work.

Four effects stand out. (i) Representation matters most: moving from PCA to UMAP is the single largest improvement. (ii) The SNN–Leiden ensemble (EMR-CGC) is the workhorse,

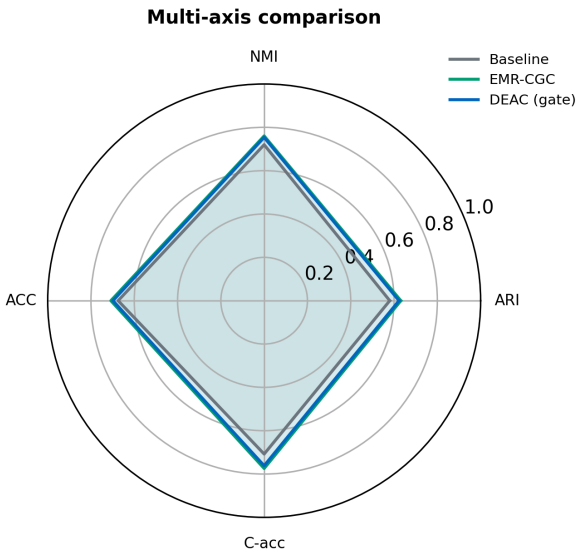


Fig. 8. Multi-axis comparison on four axes: ARI, NMI, ACC (external accuracy) and C-acc (an inverse normalised cluster-count error). The deployable gate (DEAC) tracks EMR-CGC on the external axes but is weaker on C-acc, reflecting its limited label-free cluster-count recovery.

reaching 0.631—the best deployable mean ARI in the table. (iii) The optimizer AMR-DE, used alone, falls *below* the UMAP baseline (0.563 vs. 0.580)—a direct consequence of the internal–external misalignment discussed in §V. (iv) The QualityGate's label-free arbitration does not add accuracy: it reaches 0.623, *below* EMR-CGC alone, trading a small mean-ARI cost for protection against the optimizer's worst single-engine failures. The framework's accuracy therefore comes from the ensemble; the gate is a safety mechanism, and the label-aware oracle (0.646) marks the ceiling any label-free selector would need to reach. Fig. 11 stacks these increments as a waterfall. Accuracy gains are only useful if the method remains affordable.

K. Computational Complexity

Table XIII summarizes the theoretical cost of each component.

TABLE XIII
COMPUTATIONAL COMPLEXITY OF DEAC COMPONENTS.

Component	Complexity	Typical Cost
StandardScaler	$\mathcal{O}(nd)$	linear
PCA	$\mathcal{O}(nd \cdot d')$	$\mathcal{O}(nd \cdot 40)$
UMAP	$\mathcal{O}(n^{1.14}d')$	near-linear
t-SNE (Barnes–Hut)	$\mathcal{O}(n \log n)$	near-linear*
KNN graph	$\mathcal{O}(nk \cdot d')$	$\mathcal{O}(nk \cdot 40)$
SNN graph	$\mathcal{O}(nk^2)$	$\mathcal{O}(n \cdot k^2)$
Leiden	$\mathcal{O}(m \log n)$	near-linear
AMR-DE (all evals)	$\mathcal{O}(E \cdot F_c)$	$E \approx 200$
EMR-CGC ensemble	$\mathcal{O}(M\|K\|\|R\|F_c)$	$60 \cdot F_c$
Total DEAC	$\mathcal{O}(n^{1.14}d' + nk^2)$	sub-quadratic

* t-SNE uses the Barnes–Hut approximation (output dimension ≤ 3), which is $\mathcal{O}(n \log n)$; exact t-SNE would be $\mathcal{O}(n^2)$. The sub-quadratic total therefore holds for the implemented pipeline, with the $\mathcal{O}(nk^2)$ SNN term dominating.

Measured end-to-end runtimes on an Intel i7 CPU—Wine (178) 14 s, MFeat (2000) 24 s, USPS (9298) 45 s—keep runtime practical even at the high end of our benchmarks.

V. DISCUSSION

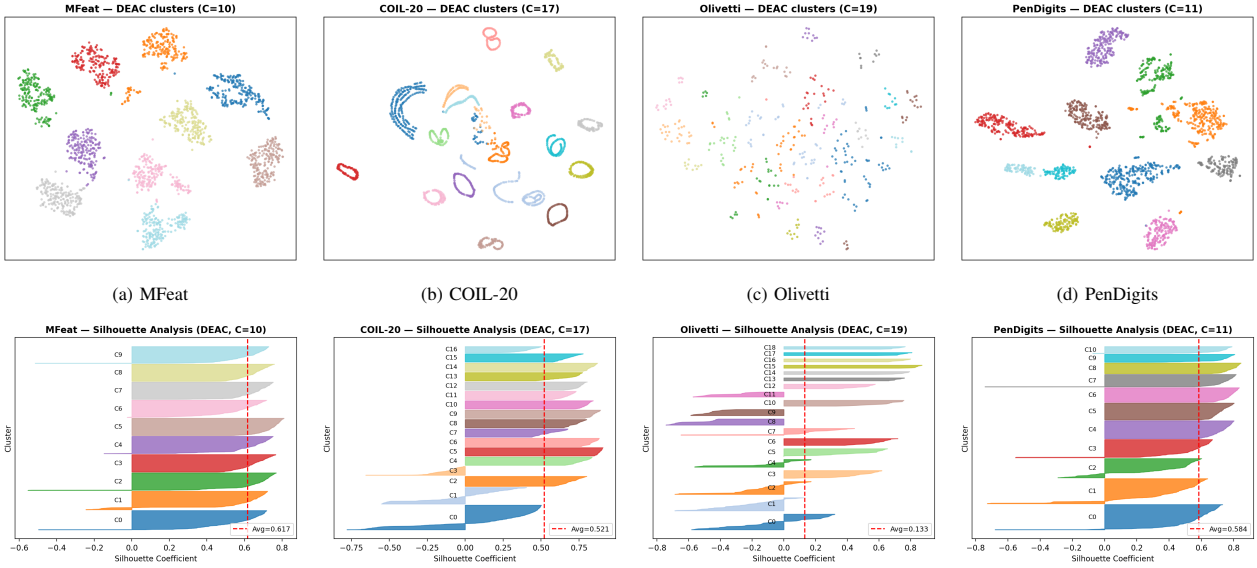
The results raise four follow-up questions: why arbitration rather than optimization, how much of the oracle a label-free gate can recover, why SNN rather than KNN, and where the method still falls short.

A. Why Arbitration, Not Optimization?

The most informative result in our study is a negative one. AMR-DE explicitly maximizes internal objectives (modularity, silhouette, Calinski–Harabasz), yet its mean ARI (0.563) falls *below* the fixed baseline (0.580) and it collapses on several datasets (Satimage 0.318, Segmentation 0.344, PenDigits 0.617). Internal quality and external agreement are therefore misaligned: a partition can be "better" by every internal score and worse by ARI. This is why DEAC does not trust any single optimized candidate; the QualityGate instead *arbitrates*, keeping an engine only when it survives three conservative, label-free tests and otherwise falling back. Crucially, the *same* misalignment that sinks AMR-DE also limits the gate: deciding by internal metrics, it reaches 0.623—*below* the ensemble it arbitrates (EMR-CGC, 0.631). Arbitration is thus a *defensive* mechanism that curbs the optimizer's worst losses, not an accuracy-maximizing one. How much of the label-aware oracle any such label-free rule can recover is the central question.

B. How Much of the Oracle Is Deployable?

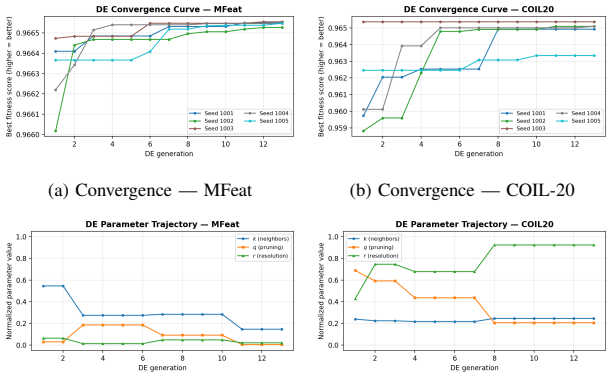
The main results (Table IV) select the best engine per dataset using ground-truth ARI—an oracle unavailable at deployment, where the gate must choose from internal metrics alone. Table XIV quantifies the gap (five seeds, full data): the deployable gate reaches mean ARI 0.623 against the oracle's 0.646 and the baseline's 0.580, recovering 65% of the oracle's gain over the baseline and agreeing with the oracle's engine choice on 6 of 12 datasets. The residual gap is a direct expression of the internal–external misalignment that motivates this work: the gate errs exactly where an engine is best by ARI but not by internal score—most visibly on Olivetti, where it prefers AMR-DE (0.237) over the stronger EMR-CGC (0.337). The same mechanism makes the gate non-monotone over the baseline—on Wine it prefers AMR-DE and loses 0.011 ARI—and, in aggregate, leaves it slightly below EMR-CGC alone (0.623 vs 0.631). We therefore treat the oracle as an upper bound and present the gate as a conservative, honestly deployable rule, and regard closing their gap—better label-free proxies for external quality—as the central open problem. The ensemble's own ingredients invite scrutiny, beginning with the SNN graph on which EMR-CGC is built.



(a) MFeat (b) COIL-20 (c) Olivetti (d) PenDigits

(e) MFeat sil. (f) COIL-20 sil. (g) Olivetti sil. (h) PenDigits sil.

Fig. 9. Qualitative analysis. Top (a–d): t-SNE projections of DEAC clusters. Bottom (e–h): silhouette plots.



(a) Convergence — MFeat (b) Convergence — COIL-20

(c) Trajectory — MFeat (d) Trajectory — COIL-20

Fig. 10. DE optimization dynamics: convergence curves (top) and parameter trajectories (bottom).

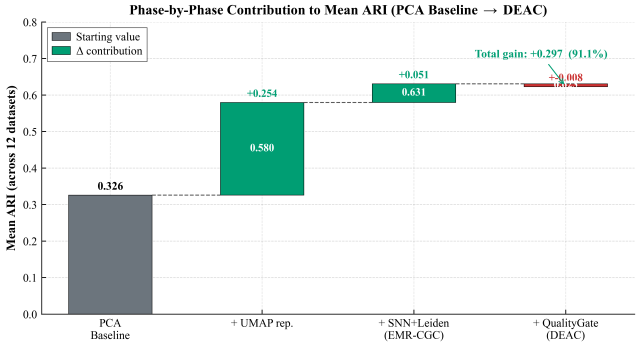


Fig. 11. Phase-by-phase contribution to mean ARI over 12 datasets. From the PCA baseline, UMAP representation and the SNN–Leiden ensemble (EMR-CGC) drive mean ARI to 0.631; the label-free QualityGate then *reduces* it slightly to 0.623 (trading accuracy for safety), while the label-aware oracle marks the 0.646 upper bound.

C. SNN vs. KNN Graph Construction

SNN weighting is not a uniform win, and it is worth being precise about where it helps. Under a matched single-graph protocol (same UMAP representation, k=35, r=0.25), swapping KNN+Louvain for SNN+Leiden changes mean ARI only marginally—from 0.580 to 0.568, a 1.2-point decrease on average. The average, however, hides a clear pattern that follows the theory of §III-F: SNN improves precisely on the *high-dimensional* datasets, where distance concentration degrades raw-distance weights—Olivetti (d=4096, +0.027), CNAE-9 (d=856, +0.028), and COIL-20 (d=1024, +0.038)—while it costs accuracy on lower-dimensional inputs (USPS −0.076, PenDigits −0.058). SNN is therefore a targeted tool for high-dimensional regimes, not a blanket improvement, and EMR-CGC's overall gain comes primarily from *ensemble diversity* (five representations × twelve configurations with best-run selection) rather than from the SNN weighting alone. Neither choice removes every limitation, which we list next.

TABLE XIV
DEPLOYABLE QUALITYGATE (INTERNAL METRICS, NO LABELS) VS. THE ORACLE UPPER BOUND, MEAN ARI OVER 5 SEEDS. Δ = ORACLE − GATE.

Dataset	Gate ARI	Oracle ARI	Δ
Wine	0.791	0.802	+0.011
Glass	0.169	0.172	+0.003
Olivetti	0.237	0.337	+0.101
CNAE-9	0.642	0.642	+0.000
COIL-20	0.754	0.789	+0.035
Optdigits	0.848	0.861	+0.013
MFeat	0.939	0.939	+0.000
Segment.	0.413	0.437	+0.024
ISOLET	0.476	0.516	+0.040
Satimage	0.525	0.541	+0.016
USPS	0.871	0.900	+0.029
PenDigits	0.811	0.819	+0.008
Mean	**0.623**	0.646	+0.023

D. Limitations

(i) *Runtime.* EMR-CGC's 60-run ensemble is costlier than single-shot methods—seconds to a few minutes per dataset in our experiments—and is the dominant cost of the pipeline. (ii) *Internal–external gap.* The gate decides on internal metrics (MOD, SIL, BAL), which do not align with ARI, so it is not strictly monotone over the baseline—on Wine it prefers AMR-DE and loses 0.011 ARI—and in aggregate scores below the EMR-CGC ensemble it arbitrates (0.623 vs 0.631); its deployable cluster-count error is also large (Table X). (iii) *Scalability.* SNN is $\mathcal{O}(nk^2)$, so very large n ($> 50{,}000$) would require approximate-NN construction. (iv) *Oracle vs. deployable gap.* The label-free gate recovers 65% of the oracle's gain over the baseline and trails N2D (Table XIV). Across learned, stability, and consensus selectors the gap does not close (best 64%), so closing it—better label-free proxies for external quality, or a learned notion of it—is the primary open problem and plausibly requires some supervision.

VI. Conclusion

We presented evidence that internal clustering metrics are an unreliable proxy for external accuracy—an engine that maximizes modularity and silhouette scores *below* an unoptimized baseline (0.563 vs 0.580 mean ARI)—and a characterization of how far a label-free rule can close the resulting oracle–deployable gap. The framework, DEAC (Defensive Ensemble Adaptive Clustering), makes this precise: a multi-representation SNN–Leiden ensemble (EMR-CGC) that infers the cluster count and is competitive with strong classical and deep baselines (0.631 mean ARI, trailing only N2D), arbitrated by a three-layer QualityGate that acts as a conservative safety net (0.623) rather than an accuracy booster. The label-aware oracle (0.646) only matches N2D, and no label-free selector we tested—metric re-weighting, learned, stability, or consensus—recovers more than 65% of its gain over the baseline, indicating the gap is structural. The lesson is that the bottleneck in automatic clustering is not clustering power but the inability to *recognize* a good partition without labels. Three extensions remain open.

The first is approximate-NN SNN construction, which would extend DEAC beyond $n{=}50$k. The second is integrating deep representation learning into the RepScan stage. The third is a semi-supervised variant that uses partial labels to close the internal–external metric gap identified in §V.

Code and data availability. The framework, all experiment scripts, and the oracle–deployable gap study are available at https://github.com/al-shaibani-star/-DEAC.

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